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Effect of Time Since Baseline Data Collection, Mechanism of Injury, and Service Academy Sub-population on Concussion Risk Prediction in Military Service Academy Cadets and Midshipmen: An Observational Study

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Abstract

Background Concussion is a common but serious health concern among athlete and military populations. To improve understanding of concussion, the NCAA and U.S. Department of Defense established the Concussion Assessment, Research, and Education (CARE) Consortium, which collected comprehensive baseline assessment data from collegiate athletes and military service academy members. In this study, we used baseline assessments collected in service academy cadets and midshipmen to estimate the likelihood of concussion in the following academic year. Using eXtreme Gradient Boosting, we determined how the predictive accuracy of baseline assessments changes over the first year after testing and compared concussion risk prediction across two sub-populations (varsity/club sport athletes and intramural cadets) and two injury mechanisms (concussions occurring during military training/physical education classes versus sports-related concussions). Exploratory analyses also identified and ranked the most influential variables contributing to risk prediction, offering insights that could guide future risk assessment and prevention efforts.

Results The analytic dataset included 16,642 participants from four U.S. military academies between the 2015–2016 and 2019–2020 academic years after data preprocessing. We used eXtreme Gradient Boosting to predict concussion risk using baseline variables collected within the same academic year. Prediction accuracy decreased as time passed since the baseline assessment both in the overall model and within each sub-population and injury mechanism model, with the overall prediction dropping from 0.68 to less than 0.64. Concussion predictions were more accurate for intramural cadets and for injuries sustained during military training or physical education, compared to varsity/club sport athletes and sports-related concussions.

Conclusions Given the significant time and resource investment required to conduct baseline concussion testing, baseline assessments being used to predict concussion risk should be performed as close to time periods of greatest risk as possible. Future research is needed to develop streamlined population- and injury mechanism-specific predictive models using a smaller set of key baseline measures to facilitate more frequent baseline testing to optimize risk prediction.

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Key Points

1. Concussion prediction accuracy decreased over the first year following baseline assessment in US service academy cadets and midshipmen.
2. Concussion prediction accuracy was greater for non-NCAA varsity or club sport service academy members and for non-sport injuries.

Introduction

Concussion, also known as mild traumatic brain injury (mTBI), is a common yet serious injury among athletes and military populations. In the United States, it is estimated that up to 3.8 million concussions occur each year due to sports and recreational activities [1]. Concussions account for 8.9% of all high school athletic injuries and 5.8% of collegiate athletic injuries [2], and these numbers may be under-reported due to athletes' reluctance to disclose symptoms or lack of awareness [3, 4]. In addition, collision sports such as rugby, American Football and ice hockey carry a higher risk of concussion than sports with lower contact [5]. Military populations also face a heightened risk of traumatic brain injury (TBI) from a variety of causes, including falls, motor vehicle accidents, and combat-related blast exposure [6]. According to the Department of Defense, mild TBI is the most frequently diagnosed form of TBI among military personnel, accounting for up to 82% of all TBI cases [7], with most of these cases occurring in non-deployed settings [8]. Cadets and midshipmen at military service academies face an elevated risk, as they are required to participate in both rigorous military training, intense athletic activities, and compulsory physical education (PE) classes that include boxing and combative training. These overlapping exposures make them an exceptionally vulnerable group.

Concussion symptoms impact both cognitive function and physical health. Symptoms of concussion can include trouble with concentration and memory, trouble with sleep, feeling overly emotional or depressed, and a number of physical symptoms, such as headache, dizziness, nausea, and sensitivity to light and sound [9]. Playing sports or engaging in physical activities with a concussion can worsen concussion symptoms, negatively impacting physical performance and increasing the risk of further injury [10]. Furthermore, individuals who have sustained a concussion are at a greater risk for subsequent concussions [11]. Repeated concussions following an initial injury are also associated with a slower recovery [12]. For military service academy members, the demands of regular training not only elevate their overall risk of concussion but also compound the potential for repeated injuries. These factors make timely identification and understanding of concussion risk critically important in this population. As such, improving methods for concussion risk assessment, early detection, and prevention is essential. Given their unique and elevated risk profile, service academy cadets and midshipmen may require specialized strategies and targeted interventions to prevent concussions.

In an effort to better understand concussions among collegiate athletes and military service academy members, the National Collegiate Athletic Association (NCAA) and the US Department of Defense (DoD) established the Concussion Assessment, Research and Education (CARE) Consortium. The CARE Consortium conducted comprehensive pre-season (baseline) assessments at participating sites, collecting detailed information on demographics, medical history, neurocognitive assessment, neurological status, postural stability, and reported symptoms [13]. We have previously leveraged the CARE database to develop predictive models to determine whether athletes and service academy members would go on to experience a clinician-diagnosed concussion during the same academic year [14].

Building on this pilot study, which used machine learning approaches to predict sports-related concussion risk across the full CARE study population using 1 year of data, we now aim to expand this work through a more comprehensive analysis within the military service academy population. In this study, we expanded our focus beyond sports-related concussions to also include injuries occurring through other mechanisms, using all years of the CARE dataset. We first evaluated how the accuracy of concussion risk prediction models using baseline assessments changes over time following the baseline assessment, with the goal of determining optimal frequency for conducting baseline assessments for concussion risk prediction. We hypothesized that prediction models would be most effective when a concussion occurs shortly after the baseline assessment, as medical and performance conditions can change over time. We then compared concussion risk prediction accuracy between service academy sub-populations and mechanisms of injury: first, between varsity/club athletes versus intramural cadets, and second, between

sports-related mechanisms of injury versus concussions sustained during military training or PE classes. We hypothesized that risk prediction models would perform more accurately in varsity/club athletes and for sports-related concussions given an expectation of more consistent and stereotyped activities in a sport setting. As an exploratory extension of our main analysis, we also examined which baseline variables were most influential in predicting concussion risk by ranking their importance applying our machine learning techniques. Comparing these key predictors across different injury mechanisms provided additional insights into how concussion risk factors may vary by context.

Methods

Study Design and Participants

This observational study utilized Concussion Assessment, Research, and Education (CARE) Consortium, data collected between the 2015–2016 and 2019–2020 academic years from four U.S. military academies, specifically the United States Military Academy (USMA), the United States Air Force Academy (USAFA), the United States Naval Academy (USNA), and the United States Coast Guard Academy (USCGA). These datasets include baseline assessments and injury records. Baseline assessments, initially repeated annually but later in the CARE study period only conducted for new cadets and midshipmen, include detailed information on demographics, sports participation, academic and family histories, as well as results from various concussion related clinical assessments. Injury records document all concussions occurring during the follow-up period, including injury date and mechanism of injury. This allows for a focused analysis of concussion occurrences and their context within this service academy population. The primary study outcome, clinician-diagnosed concussion, is defined by a consensus definition produced through evidence-based guidelines and adopted by the CARE Consortium [13, 15].

Approval was obtained from the United States Army Medical Research and Materiel Command Human Research Protection Office and the Institutional Review Board at each study site.

Variables and Data Preprocessing

We linked baseline and injury data using subject ID and examined whether each individual in the dataset sustained concussion during the same academic year (defined as May 1st to April 30th) as their baseline examination. We defined the academic year as May 1 to April 30 to capture the majority of baseline assessments at the beginning of the academic year, because most baseline assessments took place during the summer months prior to the start of the fall semester. To better understand concussion risk within our study population, we classified service academy sub-populations and injury mechanisms based on sports participation and injury context. We stratified the population based on their level of sport participation. Those participating in varsity or club sports were designated as "varsity/club athletes," reflecting their regular engagement in organized training, which may influence both their concussion risk and exposure profile. Meanwhile, those involved in intramural sports or those not reporting sport participation were classified as "intramural cadets," as they are assumed to have less frequent and less intensive sports participation. This classification enabled us to compare concussion risk and injury characteristics between groups with differing levels of athletic exposure. Similarly, injuries sustained during varsity, club, or intramural sports were classified as “sports-related concussions.” In contrast, concussions occurring during military training exercises or PE classes were classified as “military training/PE concussions.” A smaller number of cases (n = 184), coded in the dataset as “Outside of sport and military activities”, occurred as a result of other injury mechanisms not falling into either of these categories (e.g., motor vehicle accidents, falls, fights, etc.) and were excluded from the mechanism-specific analyses. Participant flow is illustrated in Fig. 1.

In the baseline dataset, the Demographics and Personal/Family Medical History unique case report form (CRF) captures demographic data, information on current and previous academic and sport history, concussion and other prior personal or family medical history, and additional social determinants of health. Neurocognitive functioning was evaluated using ImPACT [16–18], including assessment of psychomotor function, processing speed, attention, learning, memory, and working memory. Additional “Level A” assessments, which are required across all sites, include the Brief Sensation Seeking Scale (BSSS) [19], the Sport Concussion Assessment Tool (SCAT) Symptom Checklist [20], the Brief Symptom Inventory (BSI)-18 [21], the Standardized Assessment of Concussion (SAC) [22–24], and the Balance Error Scoring System (BESS) [25, 26]. In addition, optional “Level B” assessments collected at some sites include clinical reaction time [27–29], advanced postural stability measures (NeuroCom Sensory Organization Test [30]), oculomotor/oculovestibular assessments (King–Devick Test [31], Vestibular Ocular Motor Screen [32].), and/or quality of life evaluations (Short-Form Health Survey-12) [33], Hospital Anxiety and Depression Scale [34], Satisfaction with Life Scale [35]). The optional nature of Level B assessments contributes to higher rates of missing data for these variables within the dataset (Supplementary Materials Table S1). Broglio et al. provide additional

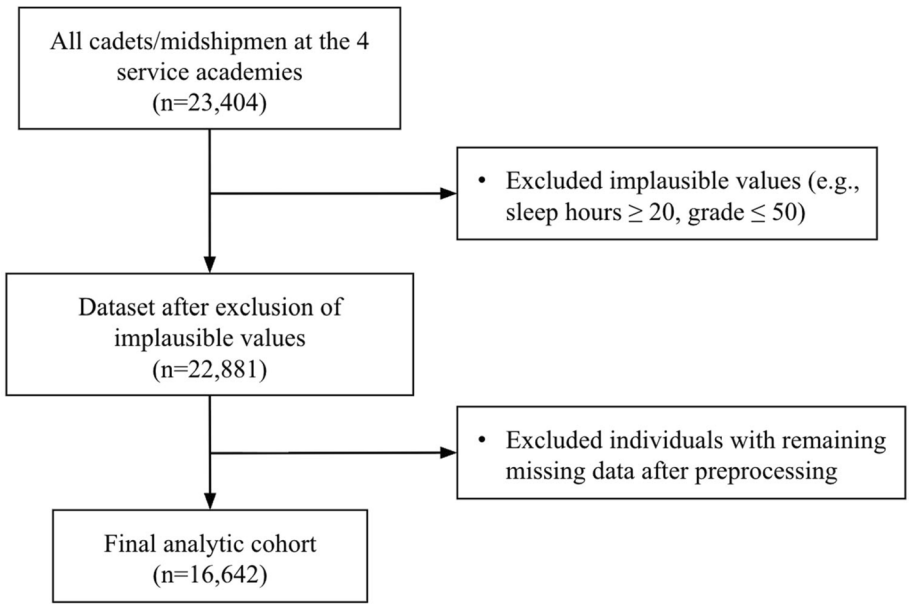


Fig. 1 Flow diagram of the data cleaning process for the analytic datasets

detailed description of the Level A and B measures with CARE Consortium Study details. [13]

We initially considered 489 variables from the above assessments (Supplementary Materials Table S1). During data cleaning, we first calculated the percentage of missingness for each variable. A total of 318 variables with more than 10% missing data were reviewed for potential consolidation or removal. To reduce missingness in key domains (e.g., sport participation, concussion history, and family medical history) we manually combined related variables. For example, in the sports-related variables (e.g., helmet use, mouthguard use), we cross-referenced participants' reported sports to infer and code the usage of protective equipment. We also recoded 133 concussion history variables based on whether participants had reported experiencing a concussion, regardless of diagnosis status. If no concussion history was reported, related entries were coded as "not applicable." We also collapsed multiple binary variables regarding concussion injury details (e.g., loss of consciousness, post-traumatic amnesia) into a single indicator reflecting whether the participant had ever experienced concussion with these features. Similarly, in the family history section, we consolidated variables referencing the same condition across different family members into a single variable. For instance, instead of 4 separate indicators for each relative's history of migraine, we created a single variable: any family history of migraine headaches. In addition, 2 variables with uniform entries or very limited variability were removed. After the cleaning process, the dataset was refined to include 87 variables, reflecting sport participation, social determinants of health, demographics, academic performance, medical history, concussion history, neurological history, psychological factors, as well as cognitive and balance test outcomes (Supplementary Materials Table S2).

For handling remaining missing data, we treated missing values in categorical variables as a separate category labeled "Missing." For numeric variables, we removed observations with missing values rather than performing manual imputation.

Data Analysis

We used eXtreme Gradient Boosting (XGBoost) [36], a binary classification model appropriate for unbalanced datasets, to predict concussion cases. While other tree-based models like random forest and linear models like support vector machine (SVM) were also considered, XGBoost provided the most powerful predictive performance. To evaluate model performance, we divided the dataset into 60% training and 40% test sets and used the area under the curve (AUC) as the primary metric [37]. AUC is a commonly used index that combines sensitivity and specificity to evaluate the overall performance of a binary classification model, in our case, service academy members with versus without concussion. Specifically, it represents the area under the Receiver Operating Characteristic (ROC) curve, which plots the true positive rate (sensitivity) against the false positive rate (1-specificity) at various threshold settings. AUC can also be described as a measure of a model's ability to distinguish between two classes.

Within each training set of the XGBoost model, we further split the data into 70% for training and 30% for validation. To make our results more reliable and to allow for later statistical comparison, we used an Out-of-Bag bootstrap method [38]. In this approach, we repeatedly created new samples from the data by randomly selecting with replacement; the data points not included in each sample (the "out-of-bag" cases) were then used as the test set. This method provided a more robust estimate of the area under the curve (AUC). We repeated this resampling process 100 times, resulting in a distribution of AUC values for further analysis. Given the class imbalance (i.e., fewer concussion cases than non-cases) we used the Synthetic Minority Over-sampling Technique (SMOTE) [39] to oversample concussion cases and down sample non-concussion cases during model training. The dataset was split into training and testing sets prior to the application of SMOTE, ensuring that synthetic data generation did not leak data into testing sets and influence the evaluation samples.

We predicted concussions occurring within a timeframe ranging from 90 days to 1 year after baseline testing and evaluated the predictive power for the overall population model as well the four sub-population and mechanism of injury models described above. To compare prediction accuracy between the two service academy sub-populations and the two injury mechanisms, we used the distributions of AUC values obtained through bootstrapping and conducted Wilcoxon rank-sum tests for statistical comparison.

As an exploratory analysis, we used XGBoost to rank each variable's contribution to concussion prediction accuracy within 90 days of baseline testing, as measured by gain. We selected a 90-day timeframe based on the hypothesis that risk prediction would be most accurate when baseline tests closely precede concussion events. After ranking the variables, we further explored the impact of the most highly predictive variables on model performance by fitting an XGBoost model with tenfold cross-validation. We iteratively assessed each model's AUC when using an increasing number of variables, starting with the most predictive and gradually increasing the number of model inputs until all variables were included. This approach allowed us to determine how well the highest-ranking variables alone could explain concussion risk compared to using the full set of variables.

Results

A total of 23,404 cadets and midshipmen participated in the study from 2015–2016 to 2019–2020. After data preprocessing, including re-coding sports-related variables, combining categories with significant missing data, and excluding incomplete numeric entries, 16,642 individuals remained in the dataset (Fig. 1). Among them, 7,851were varsity or club sport athletes and 8,735 were intramural cadets, while 56 did not indicate their sports participation level. During this time, 949 (5.7%) participants experienced concussions in various contexts: 404 during sports, 375 during military training and PE classes, and 139 outside of sports, military training activities, or PE class, while 31 did not report the injury type.

Prediction Power Over Time-Since-Baseline Testing and Between Sub-populations and Injury Mechanisms

In the prediction power plot (Fig. 2), we observed a slight decreasing trend over time following baseline tests, both in the overall model and within each sub-population and injury mechanism model, with the overall prediction dropping from 0.68 to less than 0.64. This decline occurred, even though shorter time frames resulted in fewer concussion cases, leading to more unbalanced data. Specifically, when predicting concussions within 90 days of baseline testing, the models reached an AUC of 0.65 for varsity/club athletes, 0.71 for intramural cadets, 0.67 for sports-related concussions, and 0.74 for concussions occurring during military training or PE classes. However, when predicting concussion within a year after baseline testing, the AUC values were 0.62 for varsity/club athletes, 0.64 for intramural cadets, 0.65 for sports-related concussions, and 0.69 for concussions occurring during military training or PE classes.

In addition, across all time frames, the AUC values for predicting concussion risk among intramural cadets were significantly higher than those for varsity/club athletes. Similarly, prediction of concussions occurring during military training or PE classes consistently showed higher AUC values than prediction of sports-related concussions, with both comparisons reaching statistical significance ($p < 0.001$).

Exploratory Analysis of Most Predictive Variables

Figure 3 lists the 30 most highly predictive variables for each analysis. We present only the top 30 variables here, because, as Fig. 4 demonstrates, they account for the majority of prediction accuracy. Adding more variables had less incremental effect on model performance. Of these, 25 variables were consistently important across all five models. These included seven variables related to cognitive testing, four to psychological well-being, three to balance testing, three to anthropometrics, two to social determinants of health, two to sleep, and one each related to academics, self-reported symptoms, concussion history, and demographics.

Additional sport participation variables and service academy site were important predictors for varsity/

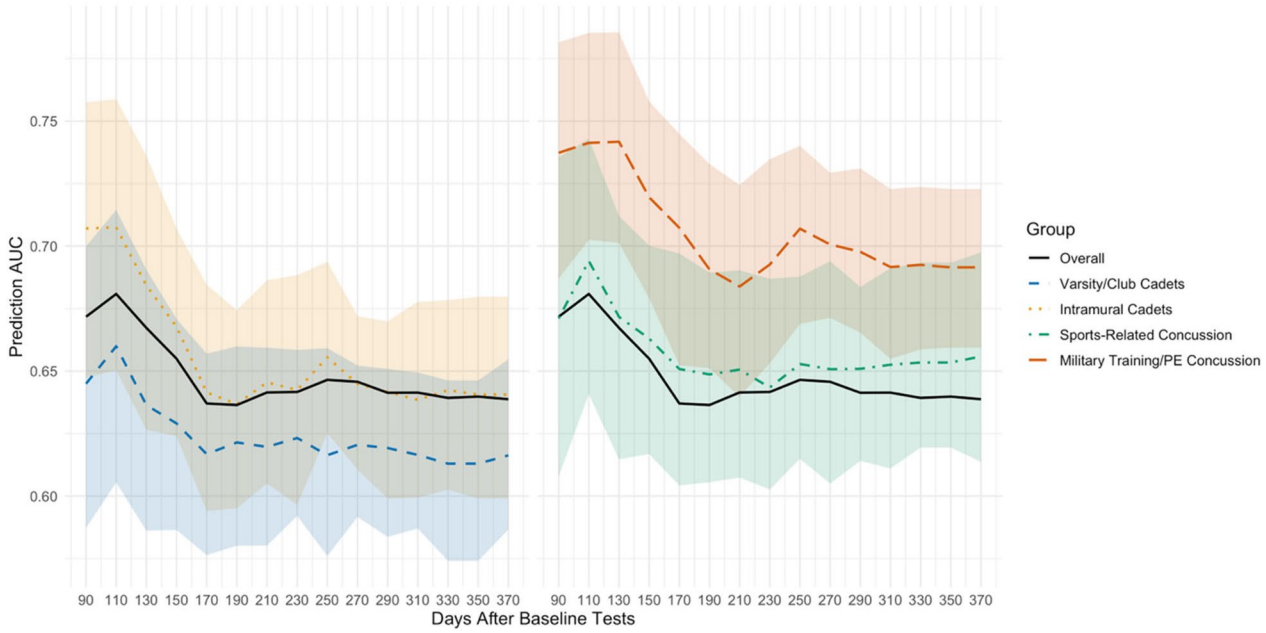


Fig. 2 Prediction power with confidence interval after baseline tests. *AUC* area under the curve, *PE* physical education

	Overall Rank	Varsity/Club Cadet Rank	Intramural Cadet Rank	Sports Concussion Rank	Military Training/PE Rank
1	Age	Number of Concussions	Weekend sleep hours	Number of Concussions	Current academic year
2	Weekend sleep hours	Age	Age	Age	Age
3	SAC total score	Weekday sleep hours	SAC total score	Sports	SAC total score
4	Number of Concussions	SAC total score	BSI18 Depression Raw Score	IMPACT memory composite - verbal	BSI18 Somatization Raw Score
5	BSI18 Somatization Raw Score	IMPACT memory composite - verbal	BSI18 Somatization Raw Score	Weekday sleep hours	BSI18 Depression Raw Score
6	IMPACT memory composite - verbal	Sports	IMPACT memory composite - verbal	SAC total score	Weekend sleep hours
7	BESS FIRM score	Weekend sleep hours	SCAT3 Symptoms Severity Score	Weekend sleep hours	BSI18 Anxiety Raw Score
8	Weekday sleep hours	BSI18 Somatization Raw Score	Weekday sleep hours	BSI18 Somatization Raw Score	IMPACT impulse control composite
9	Sports	Academic grade	BESS FIRM score	BMI	BESS FIRM score
10	IMPACT impulse control composite	IMPACT impulse control composite	BSI18 Anxiety Raw Score	IMPACT visual motor speed composite	BSSS score
11	BSI18 Depression Raw Score	BESS FIRM score	IMPACT cognitive efficiency	BESS FIRM score	IMPACT memory composite - verbal
12	BSI18 Anxiety Raw Score	Sport position category	IMPACT impulse control composite	IMPACT cognitive efficiency	SCAT3 Symptoms Severity Score
13	IMPACT visual motor speed composite	BMI	IMPACT visual motor speed composite	IMPACT impulse control composite	Weekday sleep hours
14	Area Deprivation Index by zip code	IMPACT visual motor speed composite	Number of Concussions	Area Deprivation Index by zip code	Area Deprivation Index by zip code
15	BSSS score	BESS FOAM score	Current academic year	Sport position category	SCAT3 # Symptoms Score
16	Current academic year	Height	BMI	BSSS score	Number of Concussions
17	Academic grade	Area Deprivation Index by zip code	BSSS score	Height	IMPACT cognitive efficiency
18	IMPACT cognitive efficiency	IMPACT cognitive efficiency	Area Deprivation Index by zip code	Academic grade	BMI
19	BMI	Site ID	IMPACT memory composite - visual	IMPACT memory composite - visual	IMPACT visual motor speed composite
20	BESS FOAM score	IMPACT memory composite - visual	Height	BESS FOAM score	Weight(lbs)
21	Sport position category	BSSS score	BESS FOAM score	IMPACT reaction time composite	IMPACT reaction time composite
22	IMPACT reaction time composite	BSI18 Anxiety Raw Score	IMPACT reaction time composite	BSI18 Anxiety Raw Score	IMPACT memory composite - visual
23	Height	IMPACT reaction time composite	Weight(lbs)	BSI18 Depression Raw Score	Height
24	SCAT3 # Symptoms Score	BESS total score	Academic grade	Weight(lbs)	Academic grade
25	IMPACT memory composite - visual	SCAT3 # Symptoms Score	SCAT3 # Symptoms Score	BESS total score	Sports
26	Total house income	Total house income	Total house income	SCAT3 # Symptoms Score	BESS FOAM score
27	SCAT3 Symptoms Severity Score	Weight(lbs)	Mother/guardian occupation	Site ID	Total house income
28	BESS total score	BSI18 Depression Raw Score	BESS total score	Primary position depth	Mother/guardian occupation
29	Weight(lbs)	Primary position depth	Father/guardian occupation	Mother/guardian occupation	Site ID
30	Site ID	Father/guardian occupation	BSI18 GSI Composite Score	Total house income	BESS total score

	Present in all five rankings
	Present in four rankings
	Present in three rankings
	Present in two rankings
	Present in one ranking

Fig. 3 Top 30 variables ranked from highest to lowest importance for each XGBoost prediction model

club athletes, sports-related concussions, and concussions during military training/PE classes, but were not influential in the intramural cadet model. Furthermore, variables reflecting sport position and primary position on the team's depth chart were also top predictors among varsity/club athletes and sports-related concussions, but not for among intramural cadets or military training/PE concussions.

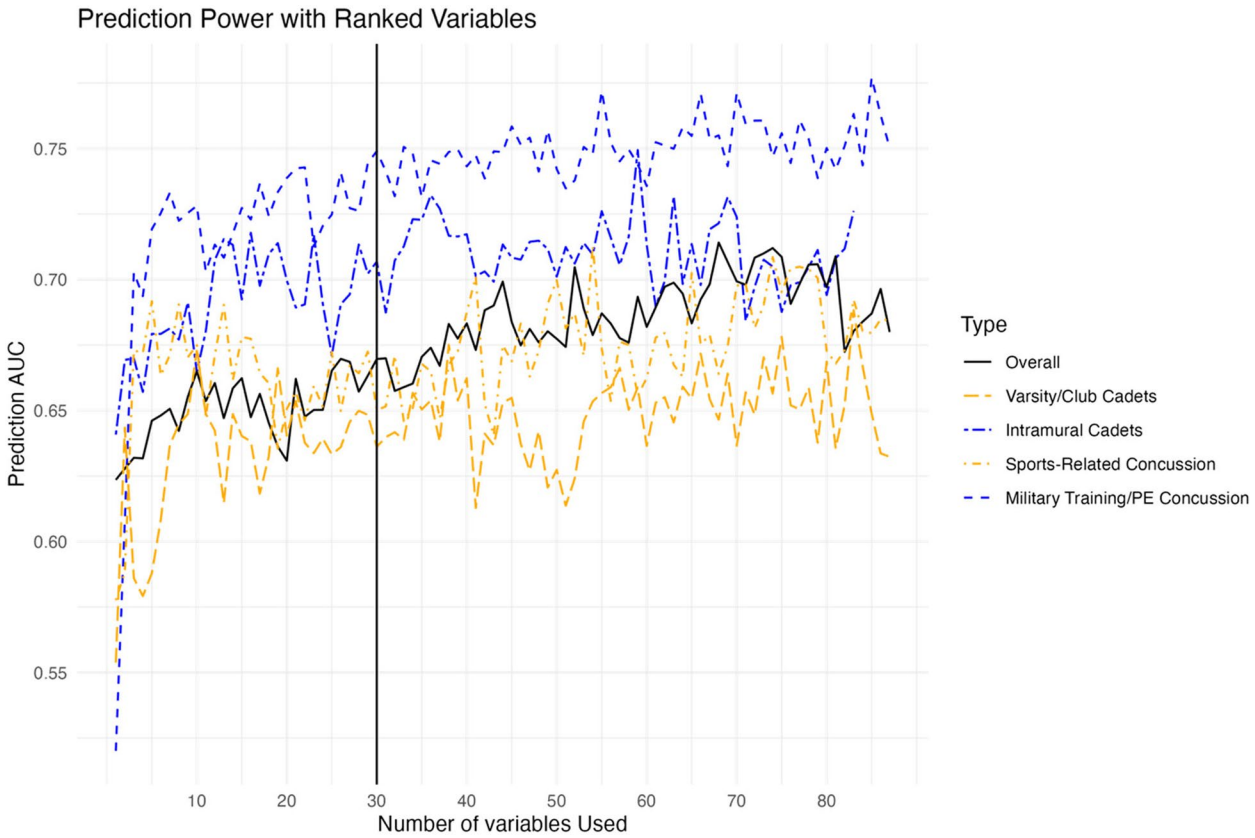


Fig. 4 Prediction power over number of ranked variables. *AUC* area under the curve, *PE* physical education

Discussion

Building on our pilot study, we initially hypothesized that it would be more straightforward to anticipate which military cadets and midshipmen might sustain a concussion if they were involved in varsity or club sports and when injuries occurred during sport. This was based on an assumption that the nature of organized sports, which are often characterized by high degrees of physical contact and clearer exposure to risk, would make it easier to link concussion risk to specific athletic participation. Conversely, predicting concussion risk among service academy members whose participation in organized sports is limited to intramural activities, was anticipated to be more challenging. We hypothesized this, because they tend to spend less time committed to organized sports and their sport participation may potentially be more varied, with less access to direct medical care on the sideline potentially obscuring identification of when and how a concussion might occur. In addition, since many intramural activities may not involve obvious high-risk play, the mechanisms leading to injury may be more diverse or subtle, at least than those typically associated with contact sports. Surprisingly, in this study, greater prediction powers were observed away from sports, specifically among cadets and midshipmen who were not involved in varsity or competitive club athletics and for non-sport injuries occurring during military training or PE classes. This finding suggests that concussions arising from non-sports or general training activities may follow more consistent or identifiable patterns than previously thought. These results highlight the importance of broadening our understanding of concussion risk factors outside organized sports settings, particularly within military populations.

Among the most predictive variables, sport played ranked as highly important both in the varsity/club athlete sub-population and in sports-related concussion predictions. Interestingly, this variable was also a key predictor for concussions occurring during military training or PE classes, but not for intramural cadets. While it is intuitive that primary sport would influence risk in sport-focused analyses, its predictive value for concussions occurring during military training and PE class suggests that the individuals' sports background may also influence their risk of sustaining concussion in these settings. This is plausible as certain sports, such as boxing, are included in military training. Service academy members who regularly participate in these sports may experience

a different risk of concussion during related training exercises compared to those without a sports background. Such a difference could potentially manifest as a higher risk due to greater exposure, or possibly a lower risk if the individuals have developed greater physical tolerance to impact exposure or represent a self-selected population due to resilience to concussion.

In practical settings, key questions often arise regarding the optimal timing and frequency of baseline concussion testing. [40] In the context of this manuscript, we are primarily interested in when baseline assessment should be conducted to optimize accuracy of concussion risk prediction. However, risk prediction has not traditionally been the primary use for baseline concussion testing. Rather, baseline results are more typically used to establish reference values to allow for direct comparisons between an individual's pre-injury health status and their condition following a suspected concussion. For this reason, some service academy members complete baseline testing well in advance of beginning their training or their sport season. In this context, a great deal of prior research has investigated the stability of baseline concussion assessment tools over time, with mixed results. For example, some studies suggest that ImPACT testing has acceptable reliability for time periods up to 2 years [18, 41], while others have reported substantially lower reliability over shorter time periods, with differences identified between individual sub-modules of the test [42, 43]. Two prior studies from the CARE Consortium found lower reliabilities over a longer retest timeframe for all components of the ImPACT test when directly comparing 1-year and 2-year retest intervals [44, 45]. In addition to time effects, concern has also been raised that baseline test results can be influenced by a number of other factors (e.g., lack of sleep [46, 47]), potentially limiting the accuracy of direct pre- and post-concussion comparisons further. In light of this, the CARE Consortium has previously concluded that repeating baseline assessments beyond an initial evaluation is unlikely to improve injury management when baseline results are being used in the traditional context of baseline-to-after-injury comparisons [44].

In this study, we extended the application of baseline assessments to include prediction of future concussion risk. This shift highlights concerns about the validity and reliability of early baseline results when they are later used to inform risk assessment, further emphasizing the need to consider timing and contextual factors in test administration. Our findings indicate that the predictive value of baseline assessments, including traditional concussion evaluation tools such as ImPACT, SAC, and BESS known to vary over time, declines as more time elapses after testing. This supports our hypothesis that baseline tests provide the most accurate information soon after the baseline test is performed and aligns with prior studies that have demonstrated greater variability between tests performed over longer test–retest intervals [44, 45]. Accordingly, these results suggest that if we were to utilize baseline assessments to predict concussion risk, at least for time-varying, non-trait measures, it would be preferable to conduct baseline assessments closer to the start of training or the sport season, rather than well in advance. In addition, our results show a notable decrease in predictive accuracy approximately 180 days, or about 6 months, after the baseline test is performed. While redoing these tests every year requires considerable time, resources, and effort, our findings raise questions about whether such an interval is sufficient for concussion risk prediction. The drop in prediction accuracy over time suggests that more frequent baseline testing may be necessary to maintain reliable assessments of concussion risk throughout the year.

While our findings indicate that more frequent baseline testing may lead to more accurate concussion risk prediction, we recognize the substantial effort and resources required to conduct these assessments with all service academy members each year. This leads us to investigate which variables contribute most to concussion risk prediction, and whether the full set of baseline assessments is necessary, versus if a more focused subset could yield similarly effective risk prediction. Figure 4 illustrates that a small number of top-ranked variables account for the majority of predictive accuracy. Models including more variables do not demonstrate substantially enhanced performance, suggesting that a relatively small set of the most predictive variables can explain the majority of an individual's concussion risk. Importantly, many of these key variables, including social determinants of health, sleep habits, academic performance, self-reported symptoms, concussion history, and demographics, can be collected through a short survey, requiring minimal resources. This approach makes annual assessments more feasible and increases the potential for practical use in clinical settings. By comparing predictive variables across service academy sub-populations and injury mechanisms, we identified both shared and unique predictors: certain variables, such as total household income, consistently ranked among the most important predictors across all models, while others were specific to particular sub-populations or mechanisms of injury. In addition to these findings, this study highlighted a significant role of social determinants of health in concussion prediction. This underscores the potential impact of broader social and environmental factors on concussion risk, beyond traditional physical or medical variables. A future research direction will be to investigate in greater detail

how social determinants, like household income, influence concussion risk prediction, including the direction of their effects on predicted risk. A deeper understanding of these relationships could help to further refine risk assessments and guide targeted prevention strategies among service academy cadets and midshipmen.

Limitations

This study predicts concussion risk among cadets and midshipmen at military service academies, and therefore, its findings may not be generalizable to civilian populations nor to other military populations. During data preprocessing, we excluded numerous variables with high rates of missing data, which may have limited our ability to identify additional predictors of concussion risk. When handling missing data, we re-coded missing values for categorical variables as a distinct "missing" category while excluding missing values for numerical variables. This approach facilitated analysis but may have introduced bias, especially if missing numerical data were non-random but related to unobserved factors.

In addition, inconsistent baseline test administration across sites and years represents another limitation. Notably, beginning in 2016–2017, the service academies modified their protocol to conduct baseline assessments only for incoming cadets and midshipmen, without repeat testing for those returning. This change may have led to data inconsistencies, since baseline tests were repeated annually before 2016–2017.

We used XGBoost as our primary modeling method. Although we considered alternative approaches, including linear models and other tree-based algorithms, XGBoost demonstrated better predictive performance. However, a limitation of XGBoost is its relative lack of interpretability, which makes it challenging to draw conclusions about individual concussion risk factors. In addition, the presence of multicollinearity among variables further complicates the interpretation, as it obscures the directionality and unique impact of each factor. In future work, we will explore strategies providing more transparent insight into the role of individual risk factors, as well as design a more streamlined model that effectively addresses multicollinearity among variables. Our ultimate objective is to create a concussion risk evaluation tool that is straightforward, clinically useful, and easy to interpret in real-world settings.

Another potential concern is that sports have different concussion risk profiles and take place at different times during the year, which may potentially influence our time-since-baseline analyses, since the majority of baseline assessments occur during the summer and early fall (Supplementary Materials Figure S1). For example, a large number of concussions occur during the fall football season. While we acknowledge potential multicollinearity between primary sport and the time interval between sport seasons and baseline testing, our stronger prediction results among intramural cadets and for injuries occurring during military training/PE class strengthen the validity of our findings, as neither are expected to be influenced by sport season. In addition, concussions occurred with a relatively uniform distribution over the year, except during the summer when sport participation and training activities are more limited (Supplementary Materials Figure S2), further suggesting against seasonality of individual sports driving our prediction results.

Finally, many baseline measures were originally developed as concussion diagnostic tools and are inherently sensitive to the effects of concussion (e.g., SCAT symptoms, SAC, ImPACT, and BESS). If individuals with lower baseline performance on these measures, i.e., baseline performance that appears more consistent with concussion, are more likely to receive a subsequent concussion diagnosis this may potentially inflate predictive associations. This concern is somewhat mitigated by the CARE Consortium's use of on-site medical and research staff to diagnose concussions without requiring threshold values be met on these measures, but caution is nonetheless warranted in interpreting these results.

Conclusion

In this study, we used baseline assessment data to predict concussion risk among cadets and midshipmen at four US military service academies. We found that concussion prediction accuracy decreases as more time elapses after the baseline assessment. Future research should investigate prediction accuracy beyond 1 year. We also found that model prediction accuracy was greater for intramural cadets and concussions occurring during military training/PE classes than for varsity/club athletes and concussions occurring during sport. We further explored the contribution of individual baseline factors to concussion risk prediction and identified the top predictors of concussion overall, as well as for each sub-population and injury mechanism. Most highly predictive variables were common across analyses, with some that were specific to sub-populations and injury mechanisms. Given that risk prediction accuracy declines over time, more frequent baseline data collection for non-trait, time-varying measures would be expected to yield more accurate concussion prediction algorithms. Further considering the substantial time and resources required for comprehensive baseline testing, identifying a smaller set of key measures that could efficiently be collected more frequently is of significant practical importance. In future work,

we will focus on the most highly predictive variables, with additional consideration of state versus trait variables and collinearity between variables, to develop a more streamlined prediction tool that maintains accuracy while reducing baseline assessment costs.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Author Contributions
Yili Wang: writing—original draft (lead); methodology (equal); formal analysis (lead); and software (lead). Kenneth L Cameron: conceptualization (supporting); funding acquisition (supporting); writing—review and editing (equal); and resources (equal); Michael McCrea: resources (equal) and writing—review and editing (equal); Thomas McAllister: resources (equal) and writing—review and editing (equal); Steven Broglio: resources (equal) and writing—review and editing (equal); Mariel Lavieri: conceptualization (lead); funding acquisition (lead); methodology (equal); formal analysis (supporting); and writing—review and editing (equal); James T Eckner: conceptualization (lead); funding acquisition (lead); methodology (equal); resources (equal); writing—review and editing (equal); project administration (lead); and supervision (lead). The CARE Consortium Investigators: resources (equal) and writing—review and editing (equal). All authors read and approved the final version of the manuscript.

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Data Availability
The CARE Consortium datasets, as well as the analysis code for this study, supporting the conclusions of this article are available in the Federal Interagency Traumatic Brain Injury Registry (FITBIR) repository at https://fitbir.nih.gov/.

Declarations

Ethics Approval and Consent to Participate
Approval was obtained from the United States Army Medical Research and Materiel Command Human Research Protection Office and the Institutional Review Board at each study site. All participants provided informed written consent prior to enrollment. All study activities were performed in accordance with the ethical standards laid down by the Declaration of Helsinki.

Consent for Publication
Not applicable.

Competing Interests
Dr Broglio has current research funding from the National Institutes of Health; Centers for Disease Control and Prevention; Department of Defense—USA Medical Research Acquisition Activity, and National Collegiate Athletic Association. He has past research funding from the National Athletic Trainers' Association Foundation; National Football League/Under Armour/GE; Simbex; and ElmindA. He has consulted for US Soccer (paid), US Cycling (unpaid), University of Calgary SHRed Concussions external advisory board (unpaid), and is the President of the Concussion in Sport Group (unpaid). He has received compensation for medico-legal litigation, and received speaker honorarium and travel reimbursements for talks given. He is co-author of "Biomechanics of Injury (3rd edition, Human Kinetics)," holds a patents on "Brain Metabolism Monitoring Through CCO Measurements Using All-Fiber-Integrated Super-Continuum Source" (U.S. US 11, 529, 091 B2) and "Organ Or Tissue Hemodynamic And Metabolic Function Measured Using Super-Continuum Light Sources" (US 12,089,945 B2), and is on the scientific advisory board for C3Logix (stock options). He is is/was on the editorial boards (all unpaid) for Journal of Athletic Training (2015 to present), Concussion (2014 to present), Athletic Training & Sports Health Care (2008 to present), British Journal of Sports Medicine (2008 to 2019). Dr. McCrea's disclosures include research funding from the following to his institution, the Medical College of Wisconsin: NIH, CDC, Abbott Laboratories, Dept. of Defense, NCAA, and NFL. No other personal or competing interests. Dr. McAllister's disclosures include concussion research grants from the NCAA, US Department of Defense, and NIH; royalties for a textbook on TBI from the American Psychiatric Association Press, Inc.; uncompensated membership on the Concussion Research Scientific Advisory Board for the Australian Rules Football Conference. Dr. Eckner has received research funding from the National Institutes of Health, the US Department of Defense, Carl Zeiss Meditec, Inc., and the Brain Injury Association of America within the past 3 years. Ms. Wang, Dr. Cameron, and Dr. Lavieri declare no conflicts of interest.

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